Advanced Embedded Systems Lab

Lab03: Arduino WiFi: MQTT

Hochschule Hamm-Lippstadt
Summer Term 2025
19.05.2025

Prof. Dr. –Ing. Ali Hayek

Organization

- Reminder: Labs and Projects submission:
  - Each team organizes the labs and project in one Github repository
  - Setup a Github repository
  - Send me an invitation (Ali.alhalabi@hshl.de)
  - Upload your project documentation to your GitHub repository.
  - **Contribution by team members will be observed.**

Organization

Strict deadlines for submissions

- Concept draft document **19.05.26 (group A and B), 27.05.26 (group C)**
- Final Documentation **01.07.26 (**Github, **or** document + Github link via email)

- Final presentation: 07.07.26
  - 20 Minutes per team.
  - Working hardware demonstration.
  - Teams present in order.

Organization

Hardware

- Each team will receive its own hardware **NEXT LAB.**
- If no Sketch exists by end of day**, no hardware**
- Each team is responsible for its own hardware
- Each team should return the hardware (without defects) after the final presentation.

Documentation:

- Use the **technical documentation template** to document your project, preferably, **replicate the document in your README file** (MD format)

Task 1 – Arduino Uno Wifi Rev 2 as MQTT Client

- In this part of the Lab you will use your Arduino Uno Wifi Rev 2 as MQTT Client for sending your Sensor Data to a MQTT Broker. (Use sensors and actuators from your own project idea)

- **Work in groups (of two or as a whole team)**. Each member sends data to a topic and one member receives all the data (multiple topics)

- You will need to you use the following libraries:
  - ArduinoMqttClient - https://github.com/arduino-libraries/ArduinoMqttClient

- Use the Tutorial:

https://docs.arduino.cc/tutorials/uno-wifi-rev2/uno-wifi-r2-mqtt-device-to-device

✓ **Hint: Try "broker.hivemq.com"**

Thank you for your attention!